Beneath the Ghost Forest: Biogeochemistry and Function in Soils as they Transition from Coastal Forest to Wetland

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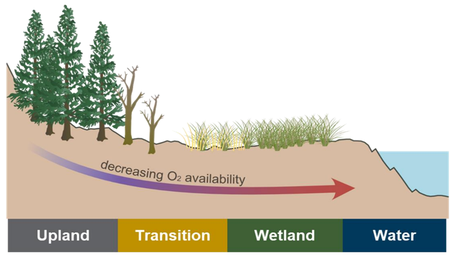
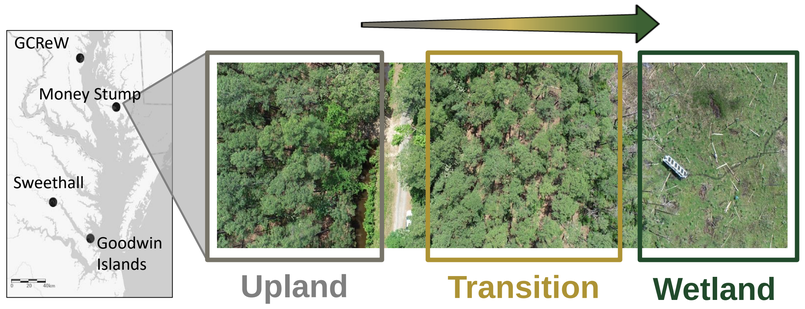
Background:

- Sea level rise drives spatial migration of coastal ecosystems and can lead to the replacement of coastal forests with tidal wetlands.
- Stressed or dead trees are well-recognized indicators of the conversion of forest to wetland, but carbon cycling and soil biogeochemistry are also altered significantly during this transition.
- Changes in soil biogeochemistry and vegetation during marsh migration may occur on different timescales and could influence ecosystem function.

Methods:

Site	Salinity
Sweethall	0-2
GCReW	8-10
MoneyStump	8-10
Goodwin Islands	18-20





- Continuous monitoring of water level and soil moisture across terrestrial-aquatic interface in the Chesapeake Bay.
- Monthly redox profiles, porewater samples, and measurements of soil greenhouse gas fluxes across upland, transition, and wetland zones.

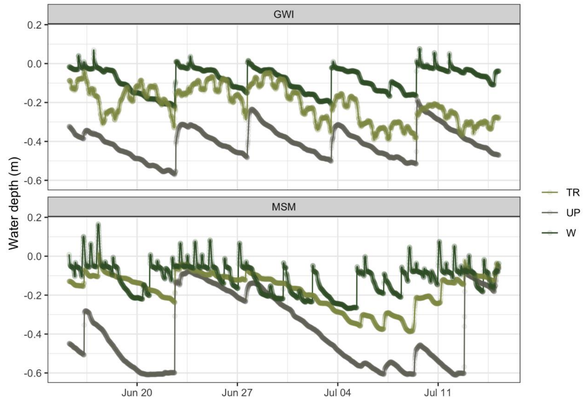


Fig.1. Water level overtime at Goodwin Islands and MoneyStump across zones.

Hypothesis: There is a hierarchical structure to spatial variation in ecosystem structure and function at the coastal terrestrial-aquatic interface set by:

1) the **depth and duration of soil saturation**
2) the abundance of **redox-sensitive elements**
3) their **interactions with plant traits**

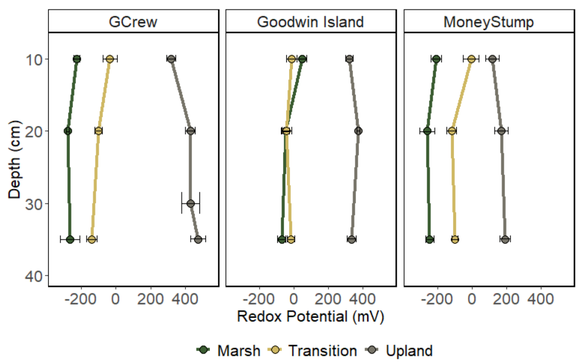


Fig.2. Redox potential profiles at GCReW, Goodwin Islands, and MoneyStump across zones.

- Lower redox potential suggests more reducing conditions
- Trends across zones align with soil moisture and water level
- Horizontal and depth gradients driven by hydrology and soil moisture

Results

- Gradient in water level across zones
- Hydrologic conditions also vary across sites
- Transition sites are highly variable
- This variability drives biogeochemical hot spots and hot moments

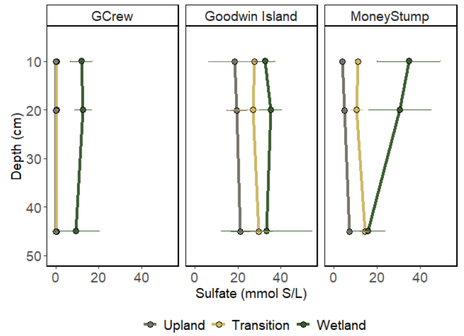


Fig.3. Porewater sulfate concentrations at GCReW, Goodwin Islands, and MoneyStump across zones.

- Porewater sulfate concentrations directly linked to salinity and conductivity
- Profiles across zones match hydrology and soil moisture

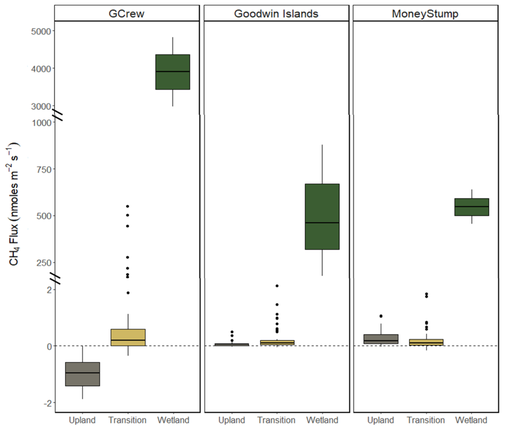


Fig.4. Soil CH4 fluxes at GCReW, Goodwin Islands, and MoneyStump across zones. Measured with Licor LI-7810 and Licor Smart Chamber.

- Soil function with respect to CH_4 is shifting prior to mass tree mortality
- Observed across site salinities and following hydrologic gradient
- Supporting that inundation may be more important to ecosystem function with respect to CH_4

Take home messages:

- Inundation depth and duration are a major drivers of structure and function at the coastal terrestrial aquatic interface
- Changes in soil biogeochemistry and vegetation during marsh migration occur on different timescales
- This asynchrony will influence ecosystem function and is critical to the framework to how we will accurately model these systems

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